

Setting up remote Linux workstation for IIC-OSIC-TOOLS

Ubuntu 24.04.4 LTS (Noble Numbat) setup

1. Download Desktop Image; we got one from [here](#)
2. Create a bootable USB stick with that image (using preferred software; we used __)
3. Restart PC and boot the Ubuntu from USB stick
4. Get through installation process with preferred software
 - we used default version
 - with 2 divided partitions (SATA Disc 240GB):
 - one for Ubuntu (100GB; /)
 - second one for software (~140GB; /home)
 - there is third one default for boot (around 1GB; /boot/uefi)
5. Partitions can be divided after the setup with the help of USB stick by booting Ubuntu from USB and dividing partitions through "Discs" manager

Remote connection to server

Here we mostly followed this [guide](#)

1. Its good to update system packages first:

```
sudo apt update && sudo apt upgrade -y
```

2. Install Vim, OpenSSH, Xrdp and XFCE. Enable ssh and change config - In 'sshd_config' file uncomment line that has Port [number] in it and change the Port number to designated one. In our case it was ports 10160-10164.

```
sudo apt install vim
sudo apt install openssh-server -y
sudo apt install xrdp
sudo apt install xfce4 xfce4-goodies -y

sudo systemctl enable ssh
sudo ufw allow ssh

cd /etc/ssh
sudo vim sshd_config
```

3. Reload system services and apply SSH configuration:

```
systemctl daemon-reload
systemctl restart ssh.socket
```

4. Configure XRDP to use XFCE and enable RDP access:

```
sudo adduser xrdp ssl-cert
echo "xfce-session" | tee ~/.xsession
sudo systemctl restart xrdp

sudo ufw allow 3389/tcp
sudo ufw enable
sudo ufw reload
```

5. In new terminal you can check if everything is working correctly; remember to logout after each login.

- '-p 10164' is port number - change to your own
- 'student' is login - change to your own

```
ssh -p 22 localhost
ssh 192.168.0.160 -p 10164
ssh 149.156.107.197 -p 10164 -l student4

ssh -L 3389:localhost:3389 student4@149.156.107.197 -p 10164
```

6. If everything is working fine from the side of the Ubuntu now you can connect from the other device.

7. First go into **Powershell** and establish the secure SSH tunnel:

```
ssh -L 3389:localhost:3389 student4@149.156.107.197 -p 10164
```

1. Open **Remote Desktop Connection** and configure settings:

- **Default/Computer:** 127.0.0.1
- **Default/Username:** student4
- **Display/Display Configuration:** set to your preferred display resolution
- **Display/Colors:** set to **True Color** or **16-bit High Color**
- **Experience/Performance:** **Automatic** or your preferred connection quality

2. Click **Connect** and log into your remote desktop.

(Optional) VSCode setup

I followed this quick [guide](#)

(Optional) Git quick setup

1. Check for updates and install git.

```
sudo apt update  
sudo apt install git -y
```

2. Set your identity.

```
git config --global user.name "Your Name"  
git config --global user.email "your.email@example.com"
```

3. Generate new ssh key.

```
ssh-keygen -t ed25519 -C "your.email@example.com"
```

4. Start the SSH agent in the background and add new key.

```
eval "$(ssh-agent -s)"  
ssh-add ~/.ssh/id_ed25519
```

5. Add ssh key to your github. Firstly copy the output of:

```
cat ~/.ssh/id_ed25519.pub
```

Create new ssh key on github website; Go into Account Settings -> SSH and GPG keys -> New SSH key. Then you can paste the `cat` output you previously copied. Everything should be set now.

6. Now you can test the connection.

```
ssh -T git@github.com
```

If it asks if you want to continue connecting, type `yes` and hit Enter. You should see a success message containing your username. Congrats, everything is working correctly.